

22PH202 – ELECTROMAGNETISM AND MODERN PHYSICS

Unit-V: ENERGY BANDS IN SOLIDS_LM3

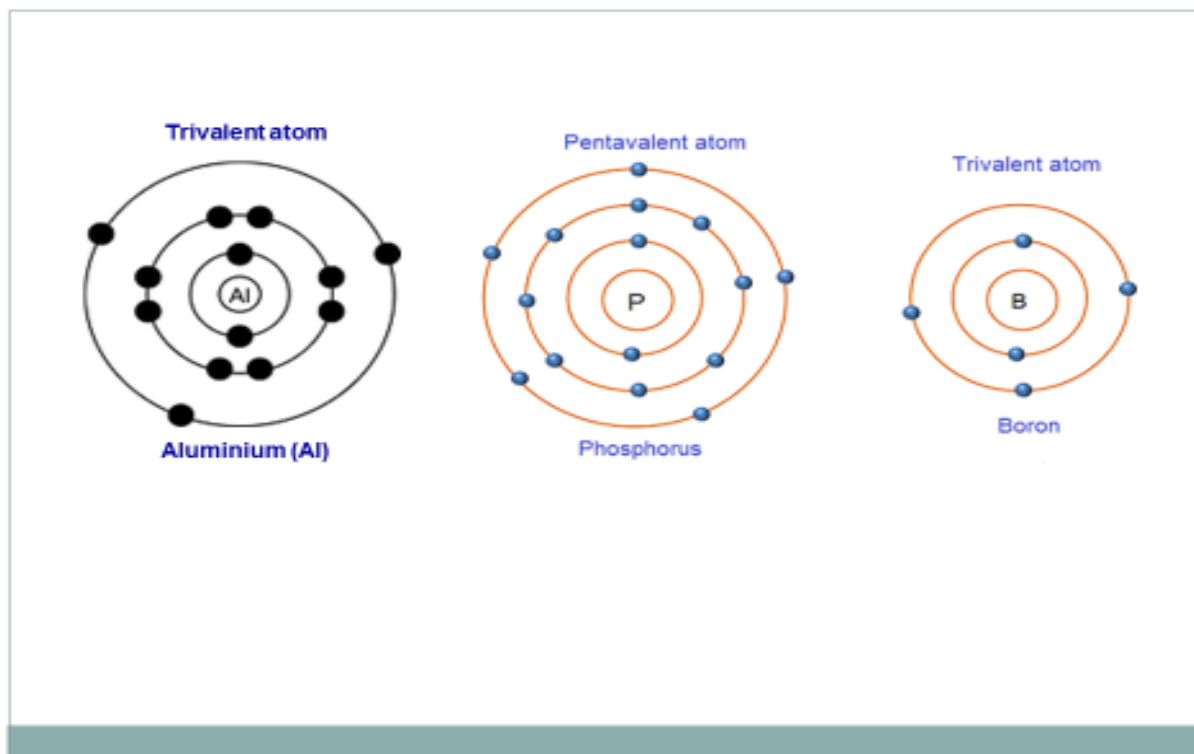
P-type Semiconductor

Trivalent Element

Element	Atomic Number
Boron (B)	5
Aluminium (Al)	13
Gallium (Ga)	31
Indium (In)	49

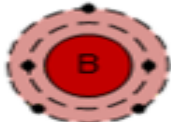
Acceptor atoms

13 IIIA 3A	14 IVA 4A	15 VA 5A
5 10.811 B Boron [He]2s ² 2p ¹	6 12.011 C Carbon [He]2s ² 2p ²	7 14.007 N Nitrogen [He]2s ² 2p ³
13 26.982 Al Aluminum [Ne]3s ² 3p ¹	14 28.086 Si Silicon [Ne]3s ² 3p ²	15 30.974 P Phosphorus [Ne]3s ² 3p ³
31 69.723 Ga Gallium [Ar]3d ¹⁰ 4s ² 4p ¹	32 72.631 Ge Germanium [Ar]3d ¹⁰ 4s ² 4p ²	33 74.922 As Arsenic [Ar]3d ¹⁰ 4s ² 4p ³
49 114.818 In Indium [Kr]4d ¹⁰ 5s ² 5p ¹	50 118.711 Sn Tin [Kr]4d ¹⁰ 5s ² 5p ²	51 121.760 Sb Antimony [Kr]4d ¹⁰ 5s ² 5p ³

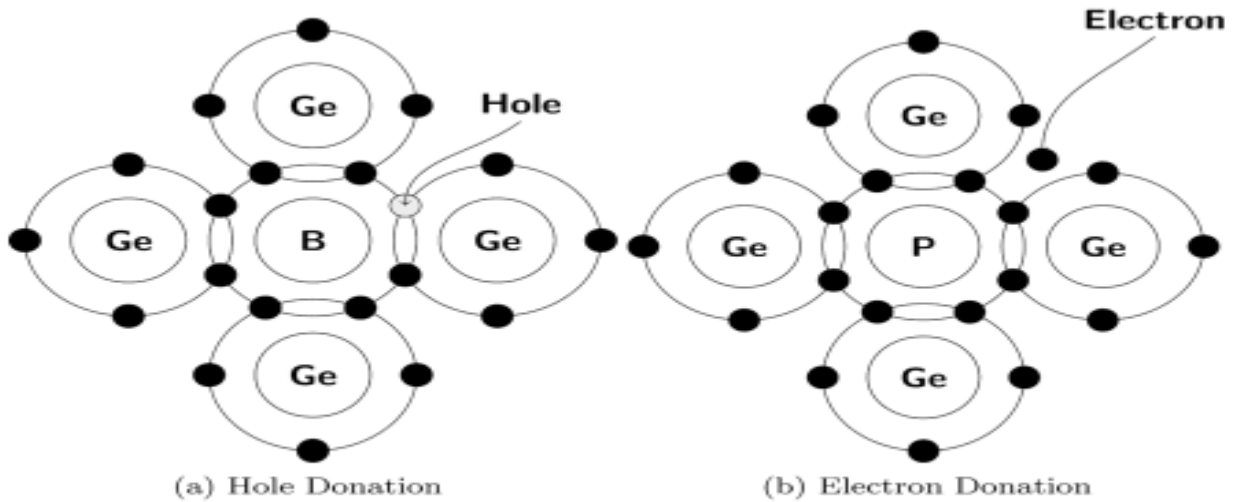
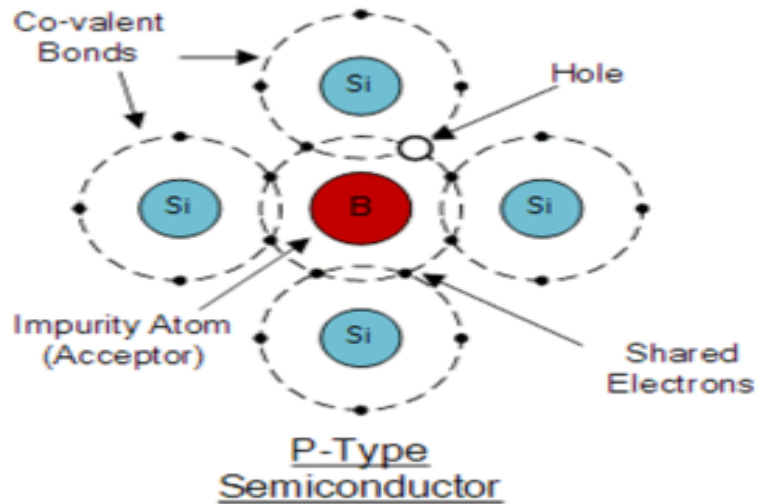


P-type Semiconductor

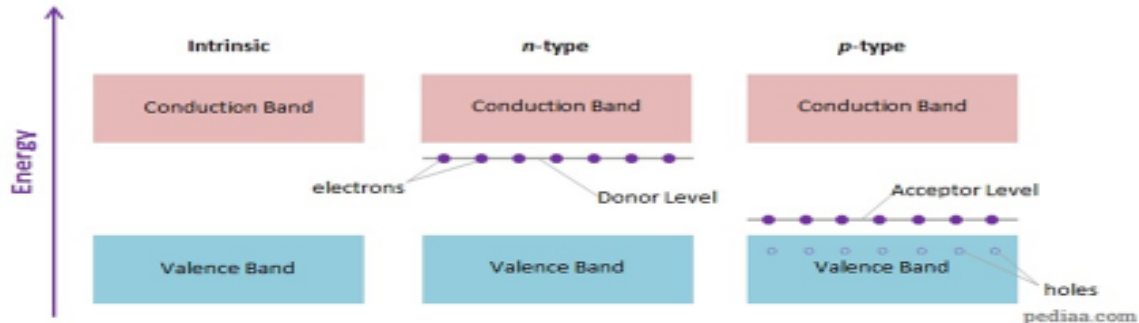
A Boron Atom,
Atomic number = "5"



Boron atom showing
3 electrons in its outer
valence shell (L)



Donor Energy Levels



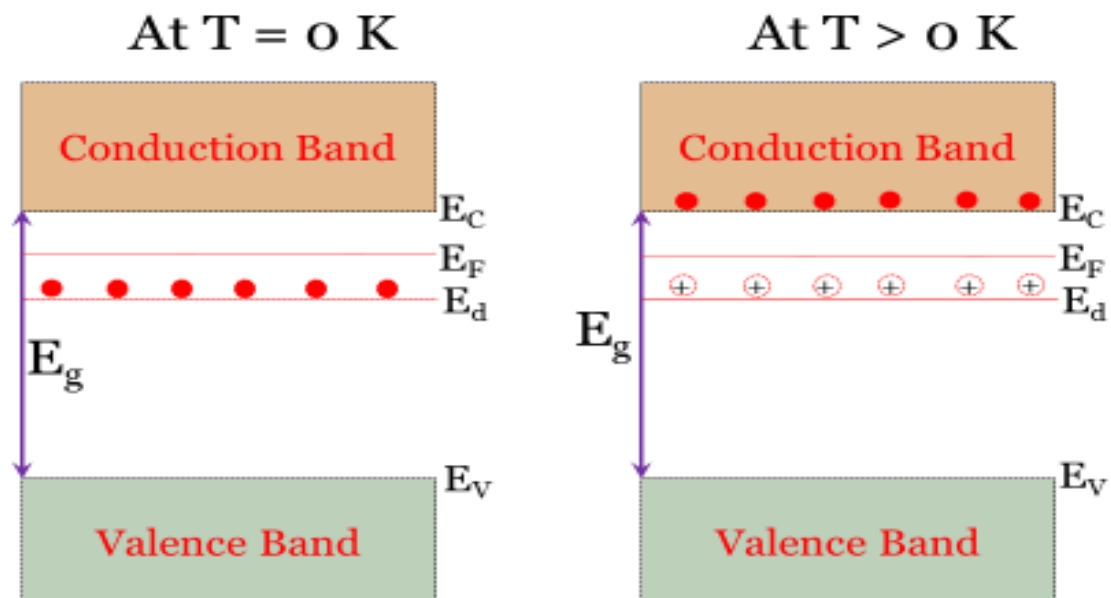
Donor level is very close to the conduction band

Most of the donor level electrons are excited into the conduction band even at room temperature (ionization of donor impurity atoms)

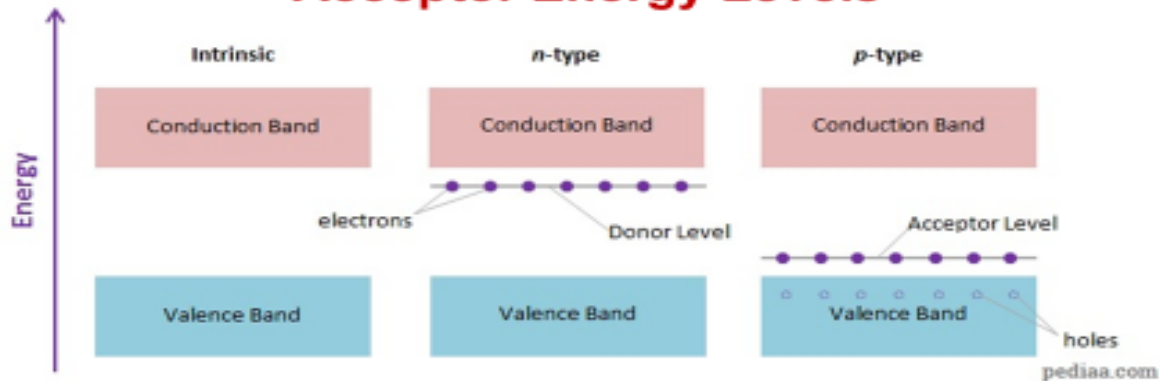
If the thermal energy is sufficient, breaking of bonds will also occur giving rise to electron-hole pairs

Electrons – Majority carriers & Holes – Minority carriers

N-type Semiconductor



Acceptor Energy Levels



Acceptor level is very close to the valence band

Even at low temperatures, acceptor atoms get ionized (take electrons from valence band and giving rise to holes in the valence band for conduction)

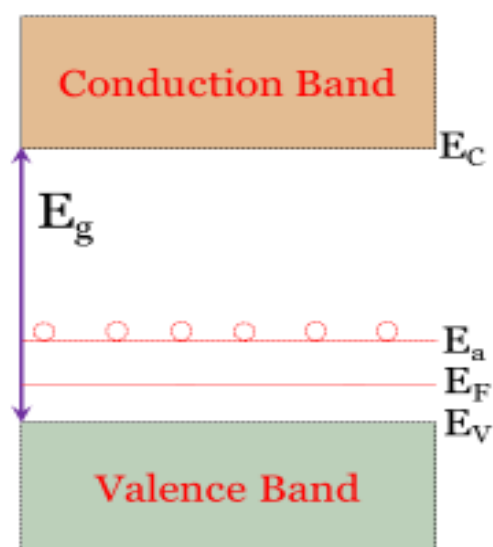
Due to ionization of acceptor atoms, only holes are created (no electrons created)

If the thermal energy is sufficient, electron-hole pairs will be generated due to breaking of covalent bonds

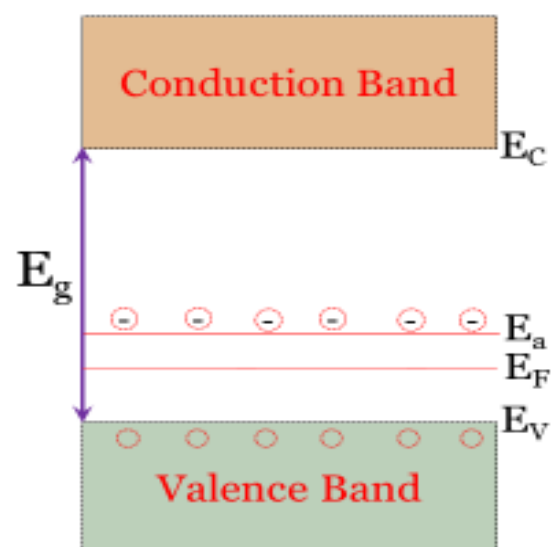
Holes – Majority carriers & Electrons – Minority carriers)

P-type Semiconductor

At $T = 0\text{ K}$



At $T > 0\text{ K}$



Applications of semiconductor devices

Semiconductor devices are all around us. . They can be found in just about every commercial product we touch, from the family car to the pocket calculator.

- **Rectifiers** which are used in d. c. power supplies.
- **Wave shaping circuits** such as clippers and clampers.
- **Voltage regulator circuits.**
- **Portable Radios and TV receivers.**
- Science and industry,
- solid-state devices, space systems, computers, and data processing equipment,
- military equipment,
- Data display systems, data processing units, computers, and aircraft guidance-control assemblies etc...